Phase 4 — Solution Architecture

Source-grounded standalone document

Solution Architecture — MVC

The source identifies an MVC architecture adapted to a headless REST API.

Layer	Responsibility
Model Layer	Defines Mongoose schemas, field rules, defaults, validation and MongoDB collection mapping.
Controller Layer	Receives requests, checks structural constraints, delegates processing and packages JSON results.
View / API Routing Layer	In the headless backend, API routes connect network paths and methods to controller handlers instead of server-rendered UI views.
Authentication Middleware	Extracts JWT tokens, validates identity context and applies authorization thresholds.
Granular Services	Enforces isolated domain rules for session lifecycles and query modifications.
Database Interface	Mongoose translates application logic into MongoDB queries.
AI Service Integration	Google Gemini SDK integration supports question answering and automated topic generation.

Request path: API Client → Express Gateway → Authentication Middleware when protected → Controller/Services → Database and/or AI → JSON response.